ECO 310: Solutions 1
Due: Thurs Feb 12

1. You are given the following partial information about a consumer's purchases in two different years (prices are per unit):

   - in year 1, when $p_1 = p_2 = \$10$, she buys 10 units of each good
   - in year 2, when $p_1 = \$10$ and $p_2 = \$7$, she buys 12 units of good 1 and x units of good 2

   For what values of x would you conclude:

   (a) That her year-1 bundle is revealed weakly preferred to the year-2 bundle? What about strictly preferred?)

   Answer: it's weakly preferred if, at the year-1 prices where she chooses it, she pays weakly (strictly) more for it:

$$\underbrace{10 \cdot 10 + 10 \cdot 10}_{\text{cost of (10,10) @ year-1 prices}} \quad \geq (>) \quad \underbrace{12 \cdot 10 + x \cdot 10}_{\text{cost of (12,x) @ year-1 prices}}$$

   So $10x + 120 \leq 200 \Leftrightarrow x \leq 8$ reveals a weak preference, and $x < 8$ reveals a strict preference.

   (b) That her year-2 bundle is revealed weakly preferred to the year bundle? What about strictly preferred?

   Answer: it's revealed weakly (strictly) preferred if, AT the year-2 prices where she chooses it, she pays weakly (strictly) extra for it:

$$\underbrace{10 \cdot 12 + 7 \cdot x}_{\text{cost of (12,x) @ year-2 prices}} \quad \geq \quad \underbrace{10 \cdot 10 + 7 \cdot 10}_{\text{cost of (10,10) @ year-2 prices}}$$

   So $170 \leq 120 + 7x \Leftrightarrow x \geq \frac{50}{7}$ reveals a weak preference, and $x > \frac{50}{7}$ reveals a strict preference.

   (c) That her behavior violates WARP?

   Answer: if (a) reveals a preference for the year-1 bundle and (b) reveals a preference for the year-2 bundle, with at least one strict. So if $x \leq 8$ and $x > \frac{50}{7}$, or $x < 8$ and $x \geq \frac{50}{7}$, or $x < 8$ and $x > \frac{50}{7}$: altogether, at least one of these happens iff $\frac{50}{7} \leq x \leq 8$. (Note that $\frac{50}{7}$ is indeed a smaller number than 8 so this range is valid).

2. Suppose Melsi (my puppy) has three toys–F (frisbee), B (ball), and S (stuffed squirrel), and always chooses one when she goes for a walk. When I only put out the frisbee and the ball, she always chooses the frisbee, i.e. $C(\{F, B\}) = \{F\}$.When it's ball or stuffed squirrel, $C(\{B, S\}) = \{B\}$. And when it's frisbee or stuffed squirrel, $C(\{F, S\}) = \{S\}$.

   (a) Explain directly that there's no hope of rationalizing these preferences, i.e. Melsi cannot possibly be making choices to maximize utility. (Hint: if she *were* rational, then saying she will only choose F from $\{F, B\}$, which is what $C(\{F, B\}) = \{F\}$ means, would imply $F \succ B$).

Amswer: if she were rational, we'd infer from these choices that

$$\begin{aligned} C(\{F,B\}) &= \{F\} \Rightarrow F \succ B \\ C(\{B,S\}) &= \{B\} \Rightarrow B \succ S \\ C(\{F,S\}) &= \{S\} \Rightarrow S \succ F \end{aligned}$$

But this preference is intransitive: $F \succ B$ and $B \succ S$ should imply $F \succ S$, and Melsi has the opposite. This contradicts our assumption that she is rational.

(b) Prove that regardless of how she chooses from $\{F, B, S\}$ (i.e. no matter what toy she grabs if I put out all three options), she violates IIA.

Answer: (1) suppose first that she chooses F; then she violates IIA by not also choosing F from $\{F, S\}$. (If it's chosen when all 3 things are available, it should also be chosen from any subset where it's an option). (2) Suppose next that she chooses B; then she violates IIA by not choosing B from $\{F, B\}$. And lastly (3) suppose she chooses S; then she violates IIA by not choosing S from $\{B, S\}$.

Since these are the only 3 possibilities, we've shown that she violates IIA no matter what.

3. (Quasilinear Preferences) A consumer's preferences over two goods are described by the following utility function:

$$u(x_1, x_2) = x_2 + \ln x_1$$

where x_1 is the amount of good 1, and x_2 is the amount of good 2. Her goal is to choose the best affordable bundle, and she has \$2 available. Quantities must be non-negative, but decimals are fine.

(a) Find the optimal bundle if $p_1 = p_2 = 1$.

Answer: here, $MU_1 = \frac{1}{x_1}$ and $MU_2 = 1$, so as $x_1 \uparrow$ and $x_2 \downarrow$, the ratio $\frac{MU_1}{MU_2} = \frac{1}{x_1}$ falls, as desired to establish convexity. So she's generally looking for an interior bundle, solving two equations:

$$\begin{aligned} \text{equal bang-per-buck} &: \quad \frac{MU_1}{MU_2} = \frac{p_1}{p_2} \Rightarrow \frac{1}{x_1} = \frac{1}{1} \Rightarrow x_1 = 1 \\ \text{spend all money} &: \quad x_1 + x_2 = 2 \end{aligned}$$

Plugging the 1st equation into the 2nd, she buys the bundle $(1, 1)$.

(b) Now find the optimal bundle if $p_1 = 1$ but $p_2 = 3$.

Answer: if you tried the above bang-per-buck method, the first equation now becomes

$$\frac{1}{x_1} = \frac{1}{3} \Rightarrow x_1 = 3$$

And then plugging this into budget line $x_1 + 3x_2 = 2$ yields $3 + 3x_2 = 2 \Rightarrow x_2 = -\frac{1}{3}$. Sadly this is infeasible ($x_2 < 0$ is not allowed), so she goes for the nearest corner: zero units of good 2 (that she wants negative), and her whole budget on good 1 (the one where she wants more than she can afford). So the bundle is $(x_1, x_2) = \left(\frac{m}{p_1}, 0\right) = (2, 0)$. (Not needed, what's happening here is that when $p_1 = 1$ and $m = 2$, she can at most afford 2 units of good 1, given nonnegativity on good 2.But $x_1 \leq 2$ implies $\frac{MU_1}{MU_2} = \frac{1}{x_1} \geq \frac{1}{2}$; so whenever $p_2 > 2$, $\frac{p_1}{p_2}$ is smaller than $\frac{1}{2}$, and so good 1 has more bang-per-buck than good 2 does, which is a nudge to spend all income on it).

4. Consider a consumer whose utility over leisure time (L) and spending money (S) is given by $u(L,S) = L^{\frac{1}{3}} S^{\frac{4}{3}}$. All spending money is earned by working at wage w per hour. There are T total hours to be divided between work time and leisure time.

   (a) Write out the budget constraint. (How much spending money is available if she wants L hours of leisure?)

   Answer: $S \leq w(T-L)$. (If she has L leisure hours, then she works $T-L$ hours at hourly wage w; she cannot spend more than this).

   (b) Use the "magic rule for Cobb-Doublas" (will be taught in Lecture 4) to argue that this consumer will choose $\frac{T}{5}$ leisure hours, regardless of the wage.

   Answer: First, rearrange the budget line in standard form with "goods" (S and L) on the LHS and "resources" on the RHS, as either

$$S + wL = wT \text{ (1) or } \frac{S}{w} + L = T \text{ (2)}$$

   Second, use a monotone transformation on the utility function to make the exponents sum to 1 instead of $\frac{5}{3}$, namely raise it to the power $\frac{3}{5}$, to obtain $L^{\frac{1}{5}} S^{\frac{4}{5}}$. So the $\frac{1}{5}$ exponent on L says that she should spend $\frac{1}{5}$ of her resource on L. With formulation (2), this directly gives $L = \frac{1}{5}T$ (this budget line is in terms of time). With formulation (1), in terms of dollars, dollar spending on L is wL and income is wT, so the rule gives $wL = \frac{1}{5}(wT) \Leftrightarrow L = \frac{1}{5}T$.

   (c) Explain why these preferences are the same as those represented by $v(L,S) = \log L + 4\log S$.

   Answer: from (b) we already took a transformation to get $L^{\frac{1}{5}} S^{\frac{4}{5}}$. Now log this to get $\frac{1}{5}\log L + \frac{4}{5}\log S$, and then multiply by 5 to get $\log L + 4\log S$. All transformations are monotone (i.e. we replaced u with $f(u)$ for some increasing function f), thus do not affect how bundles are ranked, i.e. the preference.

   (d) Using the utility function in (c), find the optimal demand bundle (L,S) as a function of w and T. Use either a Lagrangian or the intuitive bang-per-buck method.

   Answer:

   - intuitive method: we want to choose a point on the budget line that equates bang-per-buck. So

$$\frac{MU_L}{MU_S} = \frac{p_L}{p_S}$$

     But the "prices" are the coefficients on S and L in the budget lines (1) or (2), both of which give $\frac{p_L}{p_S} = w$. And differentiating utility gives $\frac{MU_L}{MU_S} = \frac{1/L}{4/S} = \frac{1}{4}\frac{S}{L}$. Equating these, we get

$$\frac{1}{4}\frac{S}{L} = w \Rightarrow S = 4wL$$

     Then plug this into either (1) or (2) (from (b)) to get $5wL = wT \Rightarrow L = \frac{T}{5}$, and plugging this back into the above, $S = \frac{4}{5}wT$. So the optimal bundle is $(S,L) = \left(\frac{4}{5}wT, \frac{1}{5}T\right)$

   - Lagrangian: the Lagrangian is

$$\mathcal{L} = \log L + 4\log S + \lambda_1 L + \lambda_2 S + \lambda_3(wT - wL - S)$$

Assuming non-binding nonnegativity constraints and a binding budget constraint, the optimality conditions are (alternatively just get rid of (5) and (6) and λ_1 and λ_2):

$$\begin{aligned}
\text{FOC for } L &: \frac{1}{L} + \lambda_1 - \lambda_3 w = 0 && (3)\\
\text{FOC for } S &: \frac{4}{S} + \lambda_2 - \lambda_3 = 0 && (4)\\
\text{non-binding constraint } L &\geq 0\text{: } L \geq 0 \text{ and } \lambda_1 = 0 && (5)\\
\text{non-binding constraint } S &\geq 0\text{: } S \geq 0 \text{ and } \lambda_2 = 0 && (6)\\
\text{binding budget} &: S + wL = wT \text{ and } \lambda_3 > 0 && (7)
\end{aligned}$$

Plugging $\lambda_1 = 0$ into (3) and $\lambda_2 = 0$ into (4),

$$\frac{1}{L} = \lambda_3 w \text{ (3a) and } \frac{4}{S} = \lambda_3 \text{ (4a)}$$

Dividing (3a) by (4a) yields

$$\frac{S}{4L} = w \Rightarrow S = 4wL$$

Plugging this into (7) yields $5wL = wT \Rightarrow L = \frac{1}{5}T$, plugging this back into $S = 4wL$ gives $S = \frac{4}{5}wT$, and finally plugging these along with $\lambda_1 = 0$ into (3) (or along with $\lambda_2 = 0$ into (4)) gives $\lambda_3 = \frac{1}{wL} = \frac{5}{wT}$ (or $\lambda_3 = \frac{4}{S} = \frac{4}{\frac{4}{5}wT} = \frac{5}{wT}$). So our solution is

$$(L, S, \lambda_1, \lambda_2, \lambda_3) = \left(\frac{T}{5}, \frac{4}{5}wT, 0, 0, \frac{5}{wT}\right)$$

These are nonnegative and satisfy all contraints, so we have a valid solution.